Md Nazim Hossain

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About Me

Passionate Computer Science student with hands-on experience in full-stack MERN development, Artificial Intelligence, and Natural Language Processing. Adaptive to applying machine learning algorithms, transformer models, and software engineering principles to solve complex real-world problems. Looking for opportunities to work at the architectural level and build useful tools.

Education

Brac University, Bachelor of Science in Computer Science and Engineering (CSE) Sept 2022 – Oct 2026

- **Overall CGPA:** 3.78/4.00 | **CSE Major CGPA:** 3.90/4.00
- **Status:** Completed 136 out of 136 Credits (Graduated: Summer 2026)
- **Key Coursework:** Data Structures, Algorithms Analysis & Design, Operating Systems, Computer Architecture, Digital Logic Design, Data Communications, Database Systems, Software Engineering, Artificial Intelligence, Compiler Design, VLSI Design, Computer Interfacing, Computer Networks, Natural Language Processing II, Introduction to Robotics.

Birshreshtha Munshi Abdur Rouf Public College, Higher Secondary Certificate (HSC) 2019 – 2021

- **GPA:** 5.00/5.00 | Group: Science

Willes Little Flower School And College, Secondary School Certificate (SSC) 2017 – 2019

- **GPA:** 5.00/5.00 | Group: Science

Skills & Technologies

- **AI & Machine Learning:** Natural Language Processing (NLP), Transformer Models (BERT, LLMs), Domain-Adaptive Pretraining (DAPT), Explainable AI (LIME, SHAP), PyTorch, Hugging Face, Scikit-learn
- **Programming Languages:** Python, C++, C, Java, JavaScript, SQL, PHP
- **Web & App Development:** Full-Stack MERN (MongoDB, Express.js, React, Node.js), React Native, Expo, HTML5, CSS3, REST APIs, MVC Architecture
- **Tools & Platforms:** Git, GitHub, Linux, MySQL, Render, Verce, OOP, Data Structures, Algorithms
- **Media & Creative:** Photography, Cinematography, Digital Photo & Video Editing

Research & Projects

NLP-Based Mental Health Detection *(Undergraduate Thesis – Brac University)* [Live Demo]

- Engineered a low-resource Bangla NLP framework achieving an **87.20% weighted F1-score** for depression severity classification.
- Designed a **decoupled LLM synthetic augmentation scheme** using Qwen3-32B during DAPT/TAPT pretraining to mitigate label noise.
- Built a **SigLIP cross-attention multimodal extension** integrating textual and visual features for holistic assessment.
- Validated model predictions using **LIME and SHAP** explainable AI frameworks to ensure model transparency and clinical trust.
- **Tools Used:** Python, PyTorch, Hugging Face Transformers, Qwen3-32B, SigLIP, DAPT/TAPT, LIME, SHAP.

Song-Lyrics Genre Classification [Github] | [Live Demo]

- Built a 3-model soft-voting transformer ensemble (BERT, RoBERTa) in PyTorch and Hugging Face to classify 53K+ lyrics across 6 genres.

- Implemented class-weighted cross-entropy loss to address severe class imbalance, achieving 0.56 Macro F1 at 0.66 accuracy.
- Deployed a live interactive inference application on Hugging Face Spaces.
- Tools Used: Python, PyTorch, Hugging Face Transformers, BERT, RoBERTa, Scikit-learn.

GardeniAR [Github]

- A cross-platform AI enriched gardening assistant application with integrated AR features to help gardeners to enjoy the art of gardening, leaving the task management, caring instructions, weather based precautions, etc. to the feature rich app.
- Tools Used: React Native with Expo and MERN Stack with MongoDB, MVC Architecture.

Meal Sync [Github]

- A dynamic meal solution for bachelors or anybody looking for home-made healthy foods on a regular subscription basis. Also, a platform for skilled home chefs to utilize their talent for secure earnings. Meal sync connects conscious consumers with talented home chefs.
- Tools Used: MERN Stack with MongoDB, MVC Architecture.

hyprGesture (Embedded Linux Gesture Controller) [Github]

- Designed and built a touchless gesture control system integrating an Arduino UNO and a PAJ7620 sensor to seamlessly control a Linux desktop environment (Hyprland) across 3 operational modes.
- Developed a continuous Python daemon running on the host machine to read, parse, and handle real-time serial data, executing system-level actions via asynchronous shell subprocesses.
- Tools Used: Python, C++, Arduino IDE, PySerial, I2C Protocol, Linux Shell Scripting, Hyprland API.

Student Help Desk [Github]

- A website for students of a University to get various help, resources, course reviews, etc. I led the team for this group project.
- Tools Used: HTML, CSS, PHP, MySQL

Drive Scraper [Github]

- A Software to scrape through a selected Google Drive folder for updates. It can dynamically send notifications in a Discord channel (Current implementation).
- Tools Used: JavaScript, Google Drive API V3, Discord Bot

Another NoteApp [Github]

- It is a note app with a beautiful UI. This app was fully deployed using render. It is built using the MERN Stack and follows MVC Architecture.
- Tools Used: MERN Stack, TailwindCSS, MVC Architecture

Experience

uSavior, Software Engineering Trainee – Dhaka, BD June 2024 – June 2025

- Participated in technical discussions evaluating modern web architectures, algorithmic approaches, and full-stack development frameworks.
- Analyzed real-world software design problems and collaborated on prototyping algorithmic solutions alongside senior developers.

General Secretary, Revolution – Youth Development, BD Oct 2022 – Now

- Encouraging youth to participate in social work to develop their personality and mindset.
- Organized and worked for several programs based around the welfare of street children.

Advisor (S-5), General Secretary (S-4), Photography Club of Rouf College – Dhaka, BD Nov 2019 – Dec 2022

- Leading a pack of enthusiasts as a Photography mentor. Managed and hosted many events while leading a team of talents.
- Overseeing the administration of the club and the club fund. Connecting students with college authorities to

encourage passionate individuals.

Co-Founder, Spread Love for Street Children - SLFC – Dhaka, BD Feb 2021 – Jul 2022

- Founding an organization with college friends to reach out to the unprivileged street children and their families in Bangladesh.
- Creating a platform for talented and passionate youths to participate in non-political social work to serve their society and nation.

Awards

- **2nd Runners Up** - CSE471 Project Showcase, Brac University.